AI for Healthcare

Assignment

Week-1

Task-1

Five interesting concepts I learned form lectures:

1) Multi-view stereo and panoramas

Cross relationship between different 2D images taken from different and algorithmically constructing the 3D physical world around them and connecting various images to get a fully interpretable image. Multi view stereo is recovering the 3D shape and panoramas is stitching various sub parts of a big image. Various applications include Robotics vision, 3D scanning and AR/VR.

2) Image classification and segmentation

Interpretation of image to classify different parts of an image. classification is assigning labels to different sub-parts of the image and segmentation is identification of pixels that belong to that particular sub-part. Together both these processes bring out information about the image in readable form. A lot of applications in healthcare, AI models/softwares ,agriculture, security etc.

3) Existence of complex eyes

Biologically there exists a lot of variety of eyes beyond simple like pin-hole eyes, mirror eyes, pit eyes, lens eyes, etc. that different beings have according to their needs and different image formation mechanisms leading to different visions. Different eyes enable different levels of acuity, focusing, different number of rods and cones enabling different color interpretation of eyes.

4) Bayer pattern and its applications

Bayer pattern is a small matrix consisting of two green, one red and one blue colour for analysing different colors of light and their different intensities of light passing through a pixel. It is widely used with CMOS sensors in most of modern day camera technologies and electronic imaging. The idea is that green color is senistive to human eyes and a large number of bayer filters are used to identify the most probable color in a particular sub-part/pixel of image.

5) Image interpolation(Resizing)

The image of estimating new pixel values when an image is transformed geometrically through processes like zooming(upscale), shrinking(downscale), rotation, warping , etc. Types include nearest neighbour interpolation, bilinear interpolation, bicubic, etc. Applications include Satellite imagery, computer imaging, scaling and many more.

Task-2

Different types of healhtcare reports:

1) A diagnostic report presents the findings of medical tests such as blood tests, X-rays, MRI scans, CT scans, and laboratory examinations. These reports help healthcare professionals identify diseases, evaluate the severity of conditions, and determine suitable treatment plans based on the test results.

2) A progress report documents a patient's health status over time during treatment or hospitalization. It includes observations, responses to medications, improvements, complications, and recommendations for further care. Such reports enable healthcare providers to assess treatment effectiveness and make necessary adjustments.

3) A discharge summary report is prepared when a patient leaves a hospital or healthcare facility. It summarizes the diagnosis, treatments provided, medications prescribed, follow-up instructions, and recommendations for future care. This report ensures continuity of care after discharge and helps other healthcare providers understand the patient's recent medical history.

4) A surgical report records details of a surgical procedure, including the reason for surgery, techniques used, findings during the operation, and postoperative outcomes. These reports serve as important legal and medical documents and assist in postoperative patient management.

Medical-imaging reports

1) X-ray scan

X-ray imaging report presents the findings from X-ray examinations of bones, chest, teeth, and other body parts. It helps identify fractures, infections, lung diseases, and abnormalities in skeletal structures. X-ray reports are among the most commonly used diagnostic imaging reports due to their speed and effectiveness.

2) CT scan

A Computed Tomography (CT) report contains observations from CT scans, which produce detailed cross-sectional images of the body using X-rays and computer processing. These reports help detect tumors, internal injuries, infections, and abnormalities in organs, blood vessels, and bones. CT reports provide more detailed information than standard X-ray reports.

3) MRI scan

A Magnetic Resonance Imaging (MRI) report describes the results of MRI scans, which use strong magnetic fields and radio waves to create detailed images of soft tissues and internal organs. MRI reports are commonly used to evaluate the brain, spinal cord, muscles, joints, and cardiovascular system. They are particularly useful for detecting tumors, neurological disorders, and soft tissue injuries.

4) Ultrasound

An Ultrasound imaging report summarizes findings from ultrasound examinations that use high-frequency sound waves to generate images of internal body structures. These reports are widely used in obstetrics, cardiology, and abdominal imaging. Ultrasound reports help assess fetal development, organ conditions, blood flow, and various soft tissue abnormalities.